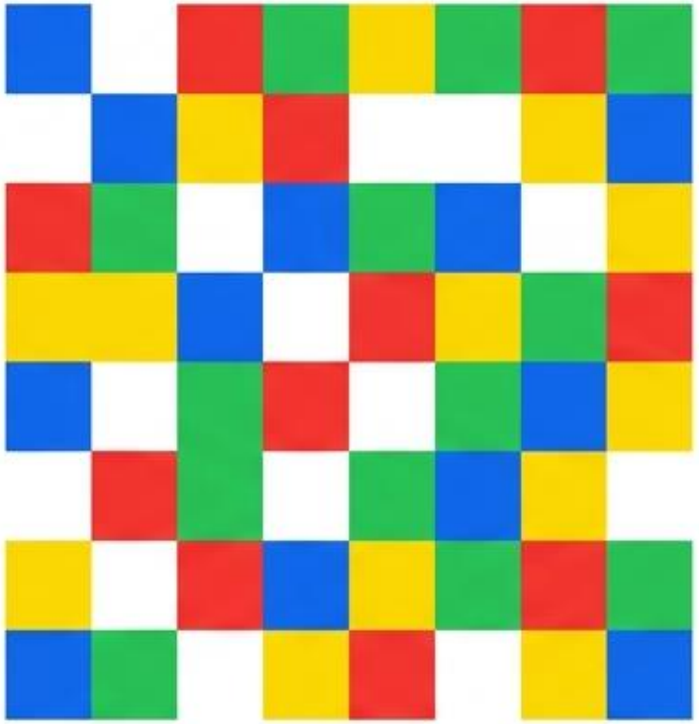


QAL: The Quantum Algorithms Lab

Visualizing the Mathematics of Quantum Computation

A virtual playground for exploring quantum algorithms, linear algebra, and group theory through direct visual interaction.

Gemini Notebook

The Primary Lab

$$\int d\alpha\, d\beta\, (\alpha|\psi\rangle\langle\varphi|\beta + \beta|\varphi\rangle\langle\psi|\alpha) \quad \sum_{i,j} c_{ij}\, |e_i\rangle\langle f_j|$$

$$\sum c_{ij}\, |e_i\rangle\langle f_j| \quad U = \exp(-iHt/\hbar) \quad \rho = \tfrac{1}{2}(I + \sigma_2)$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad \begin{pmatrix} \varepsilon_1 & \varepsilon_2 \end{pmatrix} \begin{pmatrix} 1 \end{pmatrix} \quad H|\psi\rangle = E|\psi\rangle$$

$$|\psi\rangle = \alpha|0 \qquad \rangle = E|1\rangle$$

$$|\psi\rangle = \delta|0 \qquad \rangle = E|\psi\rangle$$

$$\begin{pmatrix} \psi \\ \alpha \end{pmatrix} = \begin{pmatrix} \\ \end{pmatrix} \quad \imath i_i^{\Psi} \langle\alpha\rangle$$

$$H|\psi\rangle = \qquad \rangle = E|\psi\rangle$$

$$C = \pi\sqrt{\frac{\Omega}{g}/g} \quad \begin{pmatrix} 1 & \langle e_i \rangle & c & 0 \\ c_{ij} & \langle e_j \rangle & 0 & \langle x_i \rangle \end{pmatrix} \quad \nabla \times \Psi = \frac{\partial B}{\partial t}$$

$$\nabla \times \Psi = \frac{\partial B}{\partial t} \quad U = \exp(-iHt/\hbar) \quad \nabla \times \Psi = \frac{\partial B}{\partial t}$$

The counterintuitive nature of quantum physics makes discovering and teaching efficient algorithms extraordinarily challenging.

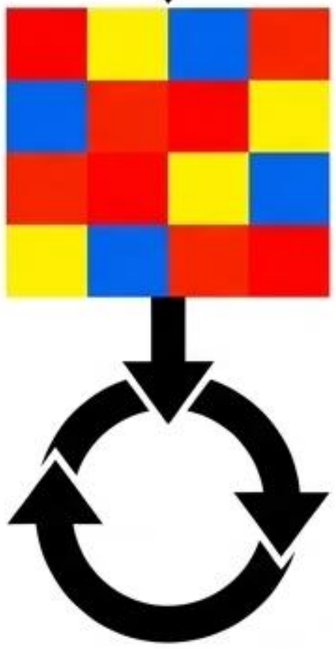


To understand the structural properties of quantum algorithms—and to find practical exponential speedups—we must move beyond layers of abstract symbols.

Gemini Notebook

The Primary Lab

$$\int d\alpha\, d\beta\, (\alpha|\psi\rangle\langle\varphi|\beta + \beta|\varphi\rangle\langle\psi|\alpha) \quad \sum_{i,j} c_{ij}\, |e_i\rangle\langle f_j|$$
$$\sum c_{ij}\, |e_i\rangle\langle f_j| \quad U = \exp(-iHt/\hbar) \quad H|\psi\rangle = E|\psi\rangle$$



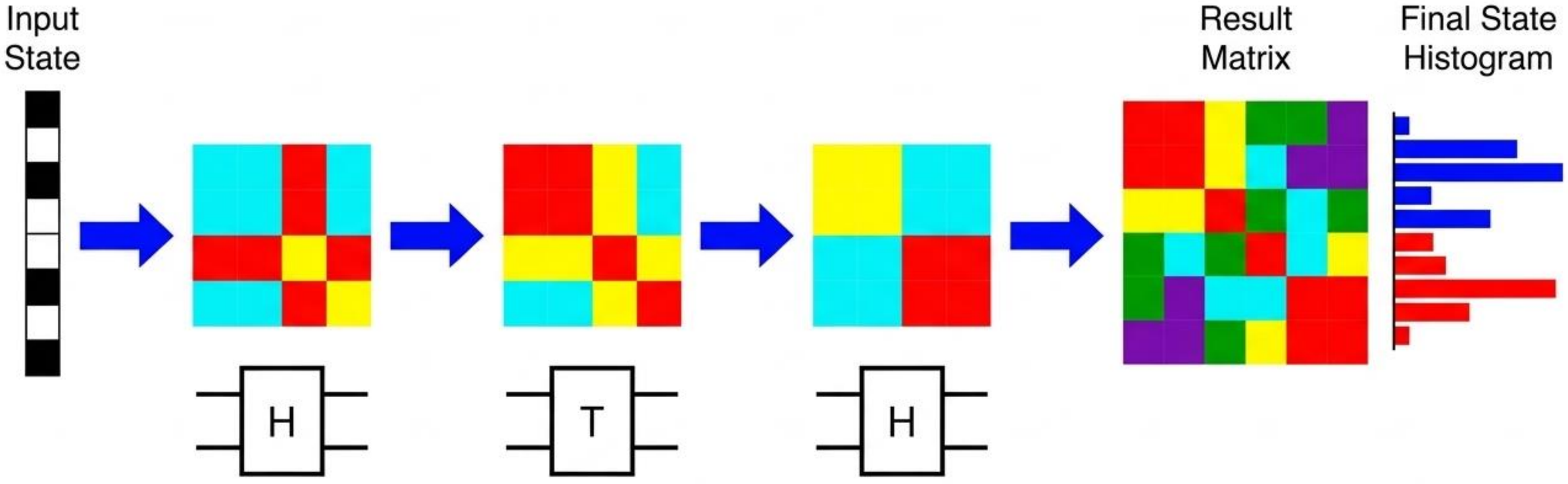
1. Textual Equations
2. Interactive Pictures
3. Experimental Mathematics

"Human brains seem to be hardwired to think in pictures. Einstein, best known for a textual equation, once said he didn't think in words at all."

Calculated Trial and Error: Making educated guesses, testing quantum operations visually, and refining them based on immediate visual feedback.

Gemini Notebook

The Primary Lab

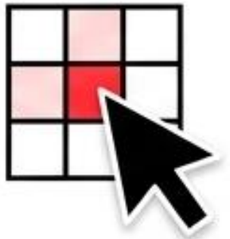


An algorithm is simply a finite sequence of quantum operations.
Mathematically, these are linear maps encoded as matrices of complex numbers.

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The Primary Lab

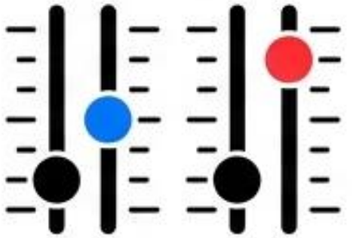
1. Select



Target single complex amplitudes, whole columns, or entire matrices natively.

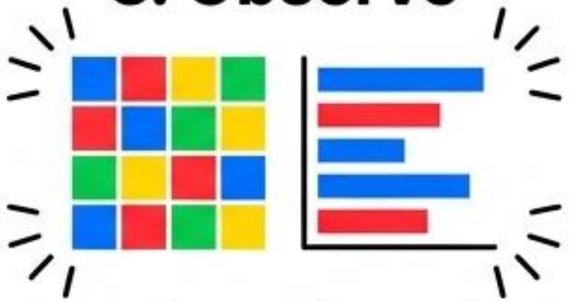
No Code Required

2. Change



Modify matrix entries directly via sliders or drag-and-drop. Zero textual or symbolic editing required.

3. Observe



Instantly see the overall operational result and final state vectors update in real-time.

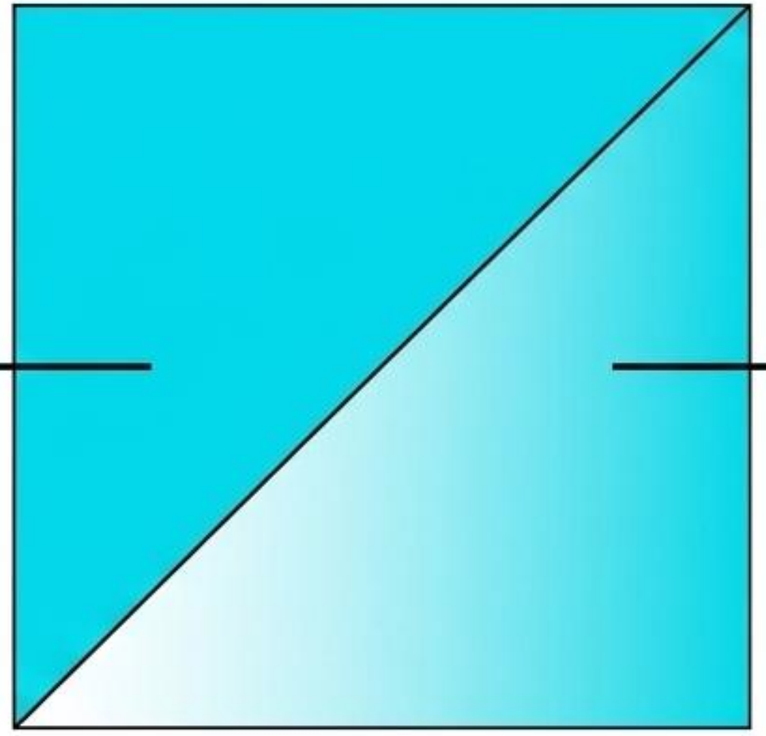
Gemini Notebook

The Primary Lab

Hue = Phase



The Hue represents the complex argument (phase/direction). Example: 0 → Red, π/2 → Yellow, π → Cyan.



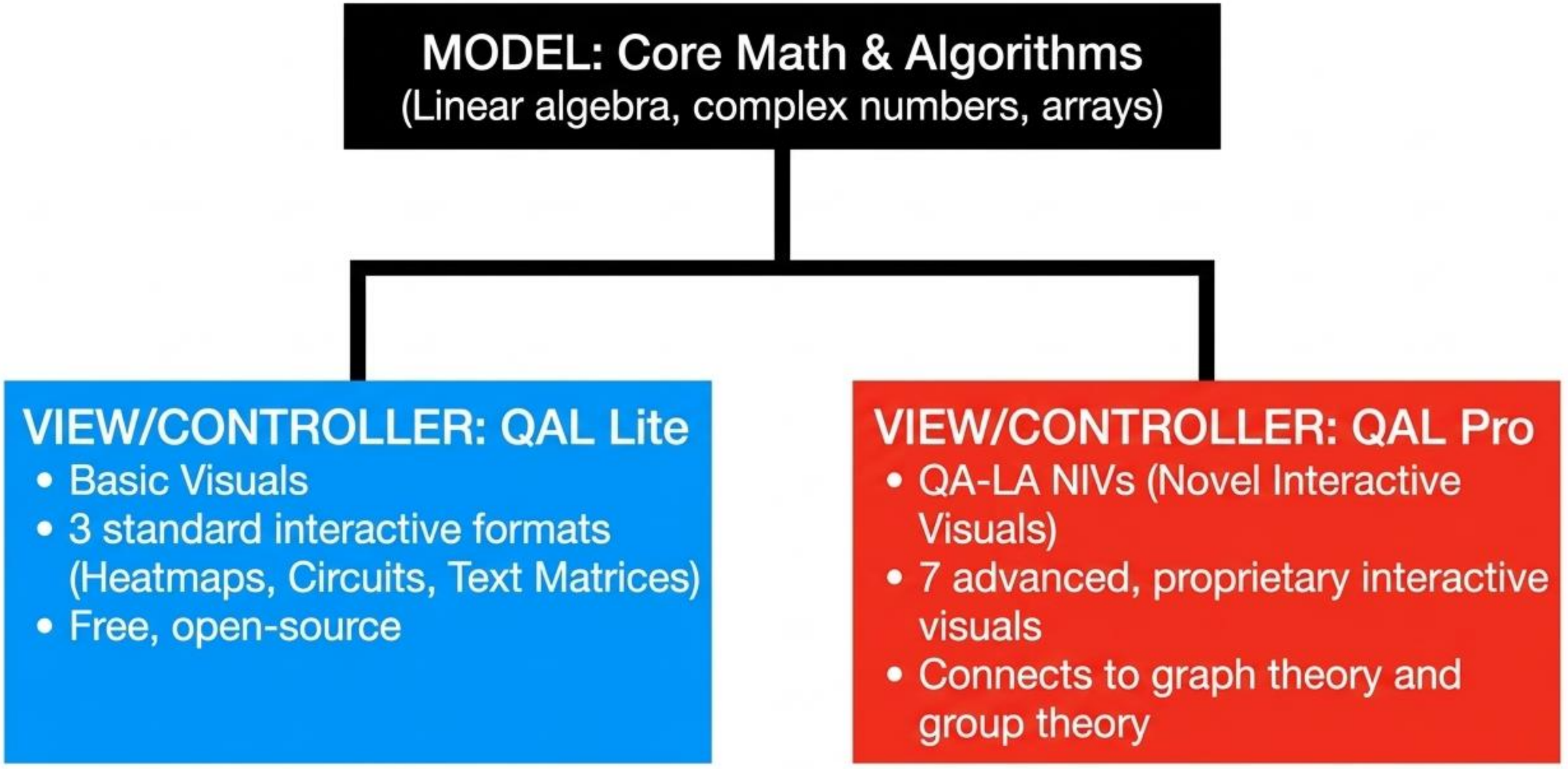
Saturation = Magnitude



The Saturation represents the complex modulus (magnitude/strength). Example: 0 (White/Empty) to 1 (Fully Saturated).

QAL translates the polar form of complex numbers directly into the brain's native language of color processing.

Gemini Notebook



Built entirely on standard web technologies (HTML5, D3.js) for universal browser accessibility without hardware bottlenecks.

Gemini Notebook

Standard Engine



QAL Lite (Basic Engine)

Employs standard heatmaps and circuits. It moves the user forward but is limited in revealing deep structural mathematics. Perfect for introductory learning.

High-performance V8 Engine



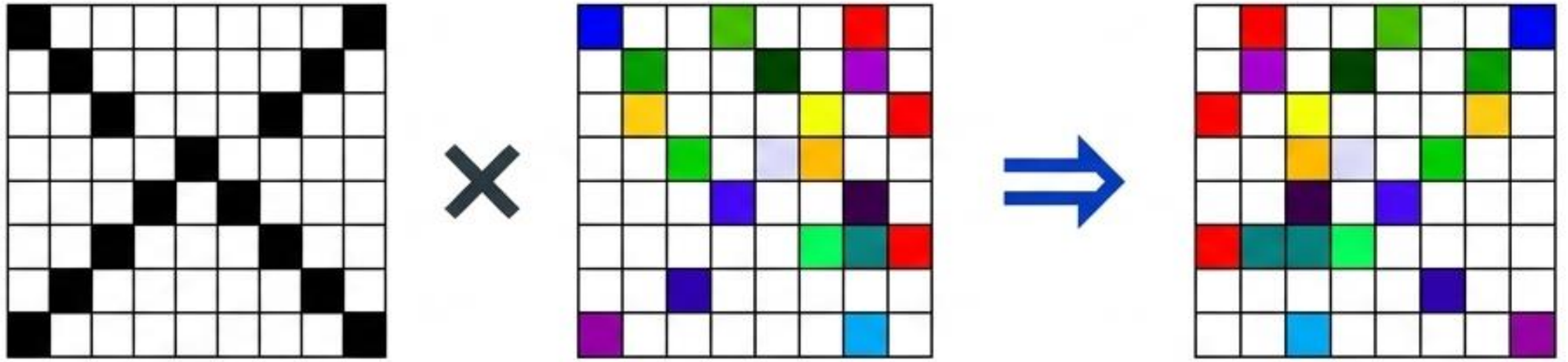
QAL Pro with QA-LA NIVs (Turbo Engine)

Unlocks 7 novel interactive visuals that expose the recursive, spectral, and group-theoretic structures of quantum operations. Extends the analytical re-reach of the researcher many folds.

Gemini Notebook

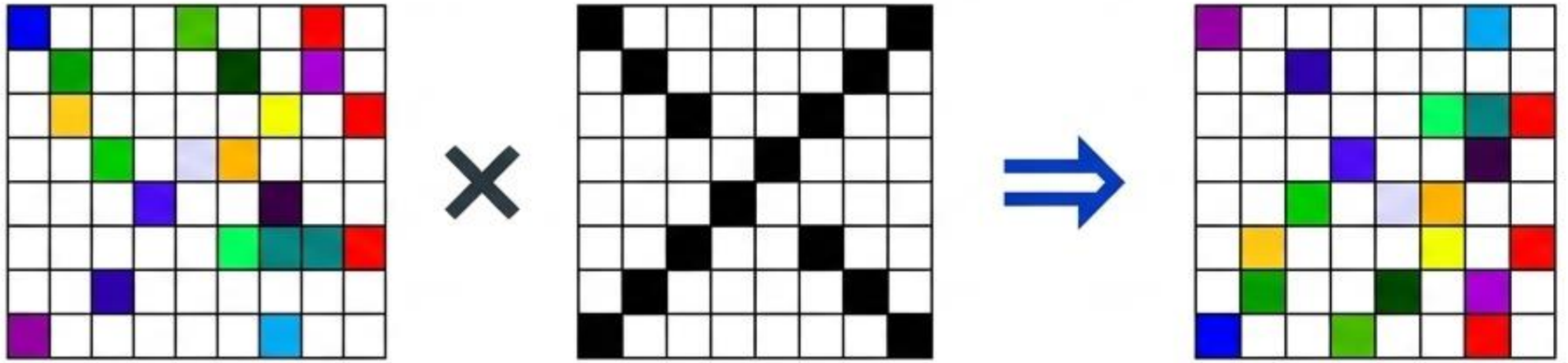
Visual Matrix Discovery: Non-Commutative Operations

Flipping Columns



Right-multiplication by X (R ∘ X) instantly reveals a left-to-right column flip.

Flipping Rows

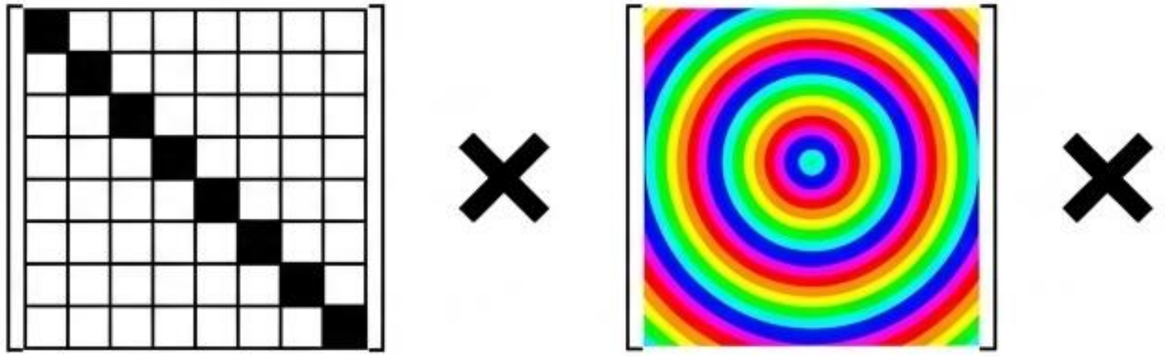


Left-multiplication by X (X ∘ R) instantly reveals an upside-down row flip.

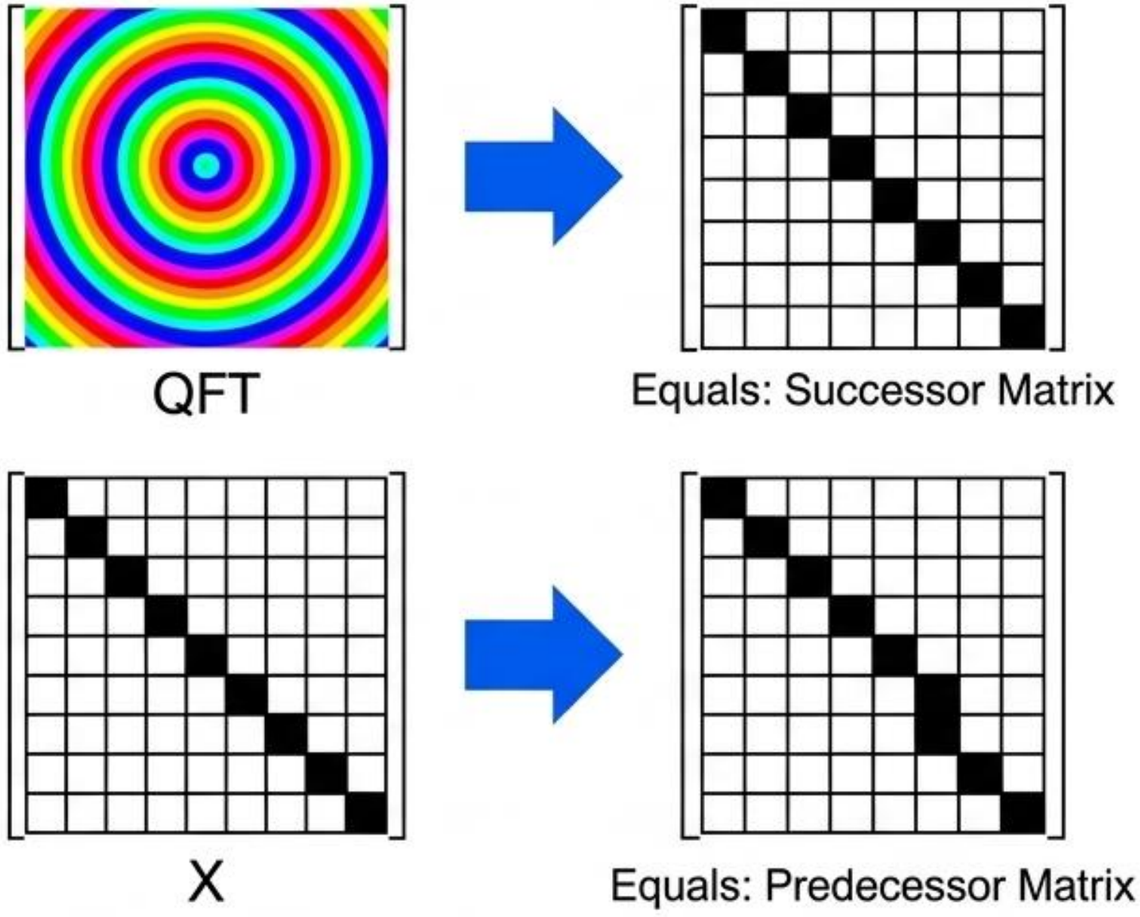
Takeaway: Non-commutative matrix multiplication becomes an instantly recognizable visual transformation rather than a tedious calculation.

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Add One / Rotate CW



Subtract One / Rotate CCW



Experimental mathematics allows users to form and verify complex conjectures purely through visual pattern recognition.

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The Abstraction Trap

Complex Numbers
↓
Abstract Vectors
↓
Unitary Matrices
↓
Sequence Multiplication

How students get lost in the math.

The QAL Solution



Rubik's Cube Cubies



Merging Paintings



Connecting Pipes

QAL translates abstract mathematical models into concrete, manipulable objects—"Digital Lego." Students interact with quantum algorithms exactly as they would physical puzzles, fostering deep, native intuition.

Gemini Notebook

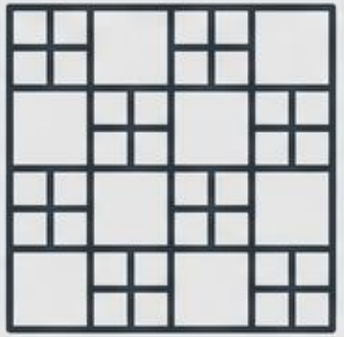
QAL for Advanced Research

Spectral Graph Theory



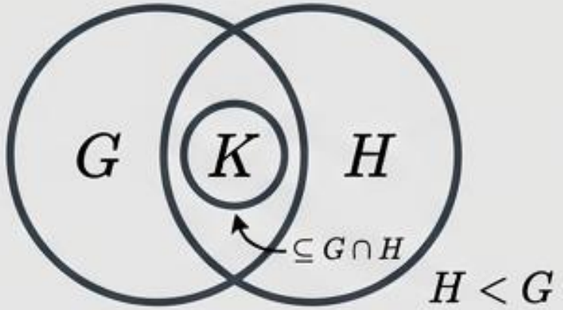
Using QAL to analyze the eigenvectors and eigenvalues (spectrum) of efficient quantum operations.

Recursive Structures



Visually exploring the self-similar, recursive nature of efficient quantum operations using tools from order and category theory.

Non-Commutativity & HSP



Investigating the non-commutative Hidden Subgroup Problem to potentially find an efficient quantum algorithm for the Graph Isomorphism (GI) problem.

Visualizing internal structure accelerates the discovery of practical quantum speedups.

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COMPETITIVE COMPARISON: QAL vs. THE FIELD

	Amplitude-Level Interaction	Zero Coding Required	Advanced Math Visualization	Real-Time Feedback Speed	Hardware Execution
QAL Pro	●	●	●	●	○
QAL Lite	●	●	○	●	○
IBM Quantum Composer	○	○	○	○	●
Quirk	○	●	○	●	○
Quantum Flytrap	○	●	○	●	○

QAL is an interactive math lab, not a hardware compiler. It fills the void between gamified learning and professional code editors.

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Human
Visual
Intuition



Rigorous
Quantum
Mathematics

Future Horizons

QAL VR
Immersive spatial linear algebra.

QAL-Lib
A visual algorithm repository.

ZX-Calculus
Native diagrammatic integration.

QAL doesn't dumb down quantum computing—it translates it into the brain's native language, making the invisible both visible and solvable.

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